Integrated Project in Computational Intelligence and IoT

Course introduction

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About this course

This Integrated Project aims at **integrating knowledge into an engineering solution**

1. **Connect** knowledge ⇝ Bring together concepts from different courses.
2. **Solve** a problem ⇝ Work with a real, open-ended problem.
3. **Make** decisions ⇝ Choose methods, technologies, and trade-offs.
4. **Build & validate** ⇝ Develop a prototype and test it with evidence.

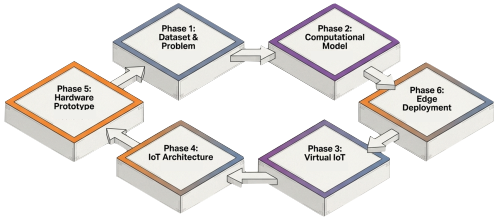
Computational Intelligence + **IoT** + **Engineering**

Intelligent IoT System

The Six Phases of the Project

From a real-world problem to a validated intelligent IoT system

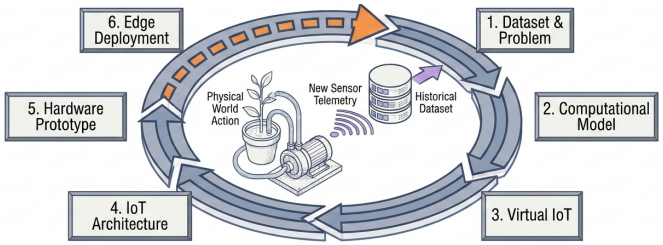
- Phase 1: Define the problem and the dataset
- Phase 2: Define the computational model ⇝ Can CI solve it?
- Phase 3: Define the Virtual IoT ⇝ Can we simulate the data flow?
- Phase 4: Define the IoT Architecture ⇝ How do we build the system?
- Phase 5: Build the Physical Prototype ⇝ Can we build it?
- Phase 6: Integration & Validation ⇝ Does it actually work?



Note: AI image generated.

Case study: Intelligent Irrigation

Can we reduce water consumption while maintaining adequate soil moisture?



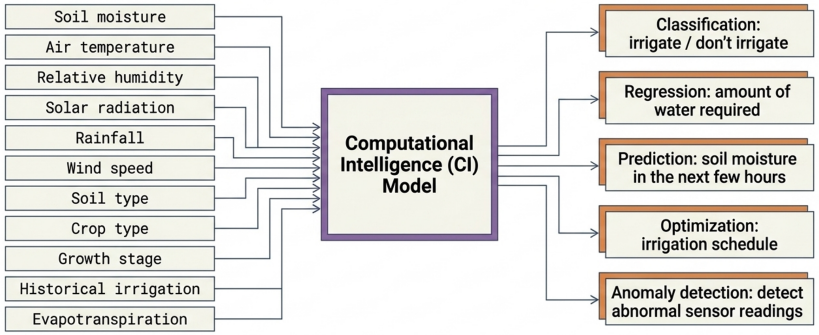
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A simple problem can lead to a complete intelligent system.

Case study: Intelligent Irrigation

Dataset

Start by defining what we need to measure, what information matters, and what irrigation decision we want the system to make.



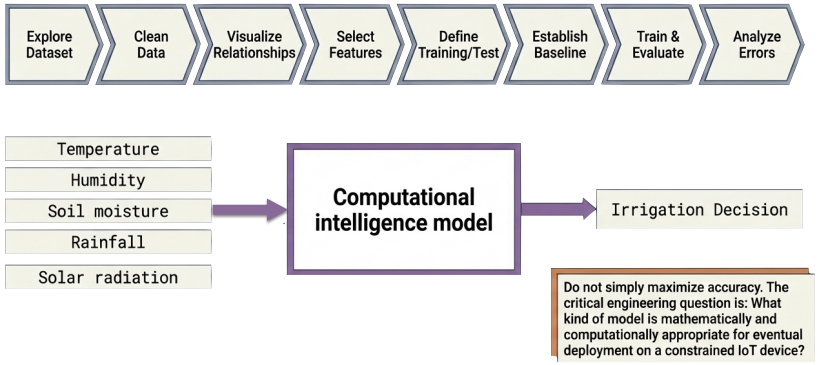
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Case study: Intelligent Irrigation

Computational model

From Dataset to Computational Model

- The model must solve a problem, not simply achieve a high score.



Note: AI image generated.

Case study: Intelligent Irrigation

From Data to Action

Can we make our model behave as if it were connected to sensors?

DATASET
Historical measurements

→

VIRTUAL SENSOR
Data replay as a sensor

→

IoT PIPELINE
MQTT
Data processing
CI model

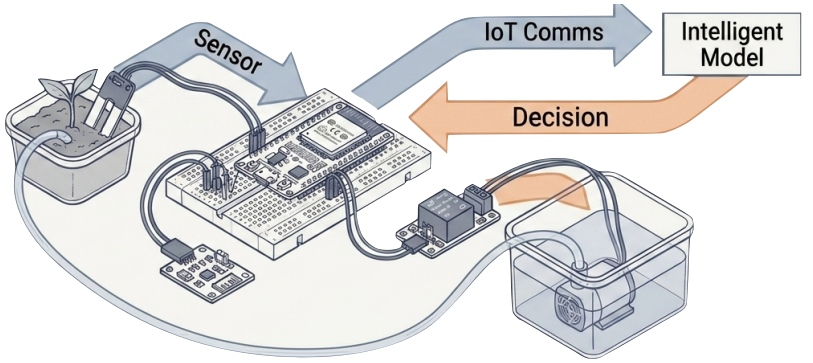
- The hardware is simulated; the data pipeline is real.
- We can test sampling rate, missing data, noise, latency, communication failures, and decision logic.

Case study: Intelligent Irrigation

Computational model

Construct the hardware prototype

- The objective is to provide the end-to-end engineering chain.



Note: AI image generated.

Course outcome

During the semester, your team will...

1. **Understand a real-word problem**
   - ⇝ Find a dataset
   - ⇝ Define the task
2. **Develop a computational model for the system**
   - ⇝ Build and evaluate the model
   - ⇝ Analyze its limitations
3. **Engineer an IoT system**
   - ⇝ Build a prototype
   - ⇝ Integrate everything

Project Milestones

M1 Problem & Dataset
- Problem definition Dataset selection Initial hypothesis

M2 First Model
- Data exploration Baseline Initial CI model

M3 Model Validation
- Experimental evaluation Error analysis Model selection

M4 IoT Architecture
- Sensors Communication Edge/cloud design

M5 Prototype
- Hardware Communication Data acquisition

M6 Final Demonstration
- Integrated system Validation Technical report

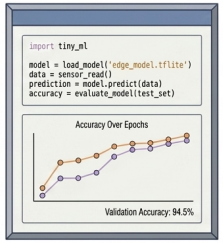
Teamwork

You will work as an engineering team.

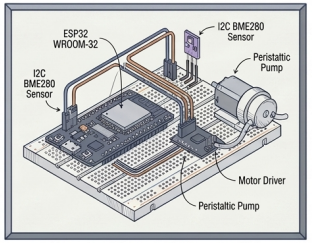
- **Shared repository**
  - ⇝ Code
  - ⇝ Data
  - ⇝ Documentation
  - ⇝ Experiments
- **Shared decisions**
  - ⇝ Architecture
  - ⇝ Model
  - ⇝ Hardware
  - ⇝ Trade-offs
- **Individual contribution**
  - ⇝ Everyone contributes
  - ⇝ Everyone understands the complete system

Bringing virtual and physical systems together

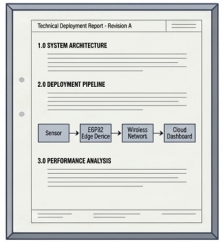
The progressive pipeline should yield a complete, engineering architecture that proves you can build intelligence and more importantly deploy it into the real word.



1. The Validated Model



2. The Physical Prototype



3. The Deployment Report

Note: AI image generated.